

WMO First Mile Data Management

World Meteorological Organization

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i. Abstract

WMO regulates the real-time international exchange of observation data to ensure the availability of high-quality observational data for many applications in particular for meteorological and hydrological forecasts, and allowing Prediction Centers (for Numerical Weather Prediction – NWP as well as climate and hydrological prediction) to be able to produce reliable products. National Meteorological and Hydrological Services (NMHSs) use NWP products to monitor and predict the weather and hydrological conditions to issue forecasts and warnings for the protection of the lives and property of citizens against natural hazards, notably in the framework of the Early Warnings for All initiative.

NMHSs operate networks comprising different observing systems to meet the observational data requirements for national, regional and international needs, and to exchange data internationally, in accordance with the WMO technical regulations. Data are collected from the observing platforms, processed and quality controlled before being used nationally and exchanged internationally. WMO does not regulate this initial data collection segment of transmission of data from the observing station to the data collection and processing system (first-mile data collection).

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

ii. Keywords

The following are keywords to be used by search engines and document catalogues.

wmo, observations, first mile, node, hub, protobuf, mqtt

iii. Security Considerations

TODO

No security considerations have been made for this standard.

Chapter 1. Scope

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This specification defines the conformance requirements for WMO First Mile Data Management (data model, encoding, transmission). Annex A defines the abstract test suite.

[1] <https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt>

[2] <https://community.wmo.int/governance/commission-membership/infcom>

Chapter 2. Conformance

Conformance with this standard shall be checked using the tests specified in Annex A (normative) of this document.

WMO First Mile Data Management defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This standard identifies numerous Requirements Classes which define the functional requirements.

The mandatory Requirements Classes for this specification are:

- "Data Model"
- "Protocol"

Chapter 3. References

- Google LLC: Protocol Buffers Documentation (2026) ^[1]

[1] <https://protobuf.dev>

Chapter 4. Terms and definitions

This document uses the terms defined in [OGC Policy Directive 49](#), which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications ([OGC 08-131r3](#)), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

The following additional terms and definitions also apply.

4.1. Node

The "thing" that collects and transmits data. Typically at the remote site, such as a weather station. This would be a device such as a datalogger or a sensor.

4.2. Host

The software at the central server that receives data from the Node and prepares it for further processing. It must be able to receive the data using the agreed Data Format and Protocol.

4.3. Protocol

The IP based set of rules that determine how data is transferred from the Node to the Host.

4.4. Data Format

A data format is the organisation of information according to specifications. It defines how the data is structured and encoded.

4.5. Data Serialization

The process of converting structured data into a Data Format that can be easily transmitted and reconstructed later.

4.6. Data Schema

A formal description in the declarative language defining the structure and constraints for the Data Format.

4.7. Observation

A single measurement data point collected by the Node, such as a temperature reading, humidity level, or wind speed.

4.8. Metadata

Additional information, such as the monitoring site ID, location, types of sensors, measured variables, measurement units, and other relevant details. It provides context for the data being transmitted.

4.9. Abbreviated terms

Table 1. Symbols and abbreviated terms

Abbreviation	Term
CF	Climate and Forecast (CF) Metadata Conventions
HMEI	HydroMeteorological and Environmental Industry
MQP	Message Queuing Protocol
MQTT	Message Queuing Telemetry Transport
PubSub	Publish / Subscribe
QoS	Quality of Service
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
WIS	WMO Information System
WMO	World Meteorological Organization

Chapter 5. Conventions

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of JSON Schema, or special notes regarding how to read the document.

5.1. Identifiers

The normative provisions in this Standard are denoted by the URI:

<http://wis.wmo.int/spec/fmdm/1>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Examples

Protocol topic examples provided in this specification are encoded as **plain text strings**.

Data and metadata message examples provided in this specification are encoded as JSON.

Complete examples can be found at <https://schemas.wmo.int/fmdm/1.0.0/examples>

5.3. Schemas

Data and metadata message schemas can be found at <https://schemas.wmo.int/fmdm/1.0.0>

5.4. Schema representation

Protobuf is used throughout this standard to define the structure of data and metadata messages.

Data and metadata message instances are always defined as JSON.

Chapter 6. Introduction

TODO

Chapter 7. Data Model

TODO one sentence

7.1. Requirements Class "Data Model"

7.1.1. Overview

This Requirements Class provides requirements for the Data Model.

Requirements Class	
http://wis.wmo.int/spec/fmdm/1/req/data-model	
Target type	Event metadata
Dependency	Protobuf
Pre-conditions	The event message conforms to the Protobuf specification.

7.1.2. FirstMessage

A FirstMessage message denotes whether the content describes data or metadata.

Requirement 1	/req/data-model/first_message
A	A FirstMessage message SHALL be used as the root definition of a message.
B	A FirstMessage message SHALL declare data or metadata .

7.1.3. Location

A Location message defines the geospatial location of a given ObserverDevice or Node.

Requirement 2	/req/data-model/location
A	A Location message SHALL define a heightMetres element in metres, relevant to the referenceSurface .
B	A Location message SHALL define a referenceSurface element using the ReferenceSurface enumeration defined in Annex B.

7.1.4. Metadata

A Metadata message defines the observers and parameters defined at a given Node. It also defines a namespace map for parameter definitions using existing controlled vocabularies.

Requirement 3	/req/data-model/metadata
A	A Metadata message SHALL define a node element using a Node message.

B	A Metadata message SHALL define one or more observers using the ObserverDevice message.
C	A Metadata message SHALL define one or more parameterDefinitions using the ParameterDefinition message.
D	A Metadata message SHALL define a namespace element using one of more maps of namespace prefixes and URIs.

7.1.5. Metadata.Node

A Node message describes the devices defines the data capture and transmits the data.

Requirement 4	/req/data-model/node
A	A Node message SHALL define a name element.

Recommendation 1	/rec/data-model/node
A	A Node message SHOULD define a location element using the Location message.

Permission 1	/per/data-model/node
A	A Node message MAY define a url element.
B	A Node message MAY define a serialNumber element.
C	A Node message MAY define a firmwareVersion element.

7.1.6. Metadata.ObserverDevice

An ObserverDevice message describes individual devices connected to Node.

Requirement 5	/req/data-model/observer_device
A	An ObserverDevice message SHALL define an id element.
B	A ObserverDevice message SHALL define a name element.

Recommendation 2	/rec/data-model/observer_device
A	An ObserverDevice message SHOULD define a location element using the Location message.

Permission 2	/per/data-model/observer_device
A	An ObserverDevice message MAY define a url element.
B	An ObserverDevice message MAY define a serialNumber element.
C	An ObserverDevice message MAY define a firmwareVersion element.

7.1.7. Metadata.ParameterDefinition

A ParameterDefinition message describes a set of variables or measurement types.

Requirement 6	/req/data-model/parameter_definition
A	A ParameterDefinition message SHALL define an id element.
B	A ParameterDefinition message SHALL define a description element.
C	A ParameterDefinition message SHALL define one or more parameters a description element.

7.1.8. Metadata.ParameterDefinition.Parameter

A Parameter message describes single variable or measurement type.

Requirement 7	/req/data-model/parameter
A	A Parameter message SHALL define a longName element.
B	A Parameter message SHALL define a unit element using the WMO Register: Code Table C-6: List of units for TDCFs . // TODO verify
C	A Parameter message SHALL define a standardNames map which describes one to many namespace prefix and URI for the equivalent parameter name in an external vocabulary.
D	A Parameter message SHALL define a device element using a DeviceRef message.
E	A Parameter message SHALL define a cellMethod element using a CellMethod enumeration.
F	A Parameter message SHALL define a cellPeriodSeconds element.

7.1.9. Metadata.ParameterDefinition.Parameter.DeviceRef

A DeviceRef message describes a reference to an ObserverDevice message.

Requirement 8	/req/data-model/device-ref
A	A DeviceRef message SHALL define an observerId element as a reference to an ObserverDevice message or a node if the ObserverDevice cannot be determined. // TODO verify

Chapter 8. Protocol

TODO one sentence

8.1. Requirements Class "Protocol"

8.1.1. Overview

This Requirements Class provides requirements for the Protocol.

Unresolved directive in sections/clause_8_protocol.adoc -
include::../requirements/requirements_class_protocol.adoc[]

Annex A: Conformance Class Abstract Test Suite (Normative)

A.1. Conformance Class: Data Model

label

<http://wis.wmo.int/spec/fmdm/1/req/data-model>

subject

Requirements Class "Data Model"

classification

Target Type:Event Metadata

A.1.1. Encoding

label

/conf/data-model/encoding

subject

/req/data-model/encoding

test-purpose

Validate that a data model is encoding as Protobuf.

TODO

TODO

A.2. Conformance Class: Protocol

Unresolved directive in sections/annex_a_ats.adoc -
include::../abstract_tests/ATS_class_protocol.adoc[]

Annex B: Schemas (Normative)

NOTE

Schema documents will only be published on schemas.wmo.int once the standard has been approved.

B.1. Data Model

```
syntax = "proto3";

package int.wmo.wis.spec.fmdm.1.conf.data_model;

import "google/protobuf/empty.proto";
import "google/protobuf/timestamp.proto";

enum CellMethod {
    CELL_METHOD_UNSPECIFIED = 0; // Required default
    POINT = 1; // Instantaneous value at a point
    SUM = 2; // Sum over the cell
    MAXIMUM = 3; // Maximum value
    MAXIMUM_ABSOLUTE_VALUE = 4; // Maximum of absolute values
    MEDIAN = 5; // Median value
    MID_RANGE = 6; // Average of maximum and minimum
    MINIMUM = 7; // Minimum value
    MINIMUM_ABSOLUTE_VALUE = 8; // Minimum of absolute values
    MEAN = 9; // Mean (average) value
    MEAN_ABSOLUTE_VALUE = 10; // Mean of absolute values
    MEAN_OF_UPPER_DECILE = 11; // Mean of the upper group of data values defined by
the upper tenth of their distribution
    MODE = 12; // Mode (most common value)
    RANGE = 13; // Absolute difference between maximum and minimum
    ROOT_MEAN_SQUARE = 14; // Root mean square (RMS)
    STANDARD_DEVIATION = 15; // Standard deviation within the cell
    SUM_OF_SQUARES = 16; // Sum of squares
    VARIANCE = 17; // Variance within the cell
}

message DeviceRef {
    oneof target {
        google.protobuf.Empty node = 1; // if field is present => refers to the single
node
        uint32 observerId = 2;
    }
}

message Parameter {
    string longName = 1;
    string unit = 2;
    map<string, string> standardNames = 3;
```

```

DeviceRef device = 4;
CellMethod cellMethod = 5;
uint32 cellPeriodSeconds = 6;
}

message ParameterDefinition {
  uint32 id = 1;
  string description = 2;
  repeated Parameter parameters = 3;
}

message Value {
  oneof kind {
    float floatValue = 1;
    double doubleValue = 2;
    int32 intValue = 3;
    uint32 unsignedIntValue = 4;
    int64 int64Value = 5;
    uint64 unsignedInt64Value = 6;
    string stringValue = 7;
    bool boolValue = 8;
    google.protobuf.Empty emptyValue = 9;
  }
}

message Observation {
  uint32 parameterDefinitionId = 1;
  google.protobuf.Timestamp time = 2;
  repeated Value values = 3;
}

enum ReferenceSurface {
  REFERENCE_SURFACE_UNSPECIFIED = 0; // Default
  MSL = 1;                          // Mean Sea Level
  GEOID = 2;                         // Geoid
  GL = 3;                           // Ground Level
  REFERENCE_ELLIPSOID = 4;          // Reference Ellipsoid
  PRESSURE_1000_HPA = 5;           // 1000 hPa Pressure Level
}

message Location {
  optional double latitude = 1;
  optional double longitude = 2;
  double heightMetres = 3;
  ReferenceSurface referenceSurface = 4;
}

message Node {
  string id = 1;
  string name = 1;
  optional Location location = 2;
}

```

```

optional string url = 3;
optional string serialNumber = 4;
optional string firmwareVersion = 5;
}

message ObserverDevice {
  uint32 id = 1;
  string name = 2;
  optional Location location = 3;
  optional string url = 4;
  optional string serialNumber = 5;
  optional string firmwareVersion = 6;
}

message Data {
  repeated Observation observations = 1;
}

message Metadata {
  Node node = 1;
  repeated ObserverDevice observers = 2;
  repeated ParameterDefinition parameterDefinitions = 3;
  map<string, string> namespaces = 4;
}

message FirstMileMessage {
  oneof content {
    // MUST be published with retained flag
    Metadata metadata = 1;

    // MUST NOT be published with retained flag
    Data data = 2;
  }
}

```

B.2. Enumerations

B.2.1. CellMethod

Name	Value	Description
CELL_METHOD_UNSPECIFIED	0	Required default
POINT	1	Instantaneous value at a point
SUM	2	Sum over the cell
MAXIMUM	3	Maximum value
MAXIMUM_ABSOLUTE_VALUE	4	Maximum of absolute values
MEDIAN	5	Median value

Name	Value	Description
MID_RANGE	6	Average of maximum and minimum
MINIMUM	7	Minimum value
MINIMUM_ABSOLUTE_VALUE	8	Minimum of absolute values
MEAN	9	Mean (average) value
MEAN_ABSOLUTE_VALUE	10	Mean of absolute values
MEAN_OF_UPPER_DECILE	11	Mean of the upper group of data values defined by the upper tenth of their distribution
MODE	12	Mode (most common value)
RANGE	13	Absolute difference between maximum and minimum
ROOT_MEAN_SQUARE	14	Root mean square (RMS)
STANDARD_DEVIATION	15	Standard deviation within the cell
SUM_OF_SQUARES	16	Sum of squares
VARIANCE	17	Variance within the cell

B.2.2. ReferenceSurface

Name	Value	Description
REFERENCE_SURFACE_UNSPECIFIED	0	Default
MSL	1	Mean Sea Level
GEOID	2	Geoid
GL	3	Ground Level
REFERENCE_ELLIPSOID	4	Reference Ellipsoid
PRESSURE_1000_HPA	5	1000 hPa Pressure Level

Annex C: Examples (Informative)

C.1. Transmission topics

C.2. Data and metadata messages

Annex D: Bibliography

- IANA: Link Relation Types, <https://www.iana.org/assignments/link-relations/link-relations.xml>

Annex E: Revision History

Date	Release	Editor	Primary clauses modified	Description
2026-06-15	Template	Tom Kralidis	all	initial revision